

SMART CAMPUS MANAGEMENT SYSTEM

SCMS software project showcase for HITEC University Taxila

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Course: Object Oriented Programming
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HITEC University Taxila



SCMS turns campus operations into an OOP demonstration.

A complete C++17 system organized into six academic modules.

1 Purpose

Show how a university can manage people, courses, library items, finance records, hostel rooms, and reports through one console application.

2 Objectives

Demonstrate inheritance, polymorphism, templates, exceptions, persistence, smart pointers, lambdas, and STL algorithms.

3 Scope

Six integrated modules plus CSV/TXT persistence, documentation, UML artifacts, executable build, and submission packaging.

6

PROJECT MODULES

25+

OOP CHECKPOINTS

C++17

COMPILER TARGET

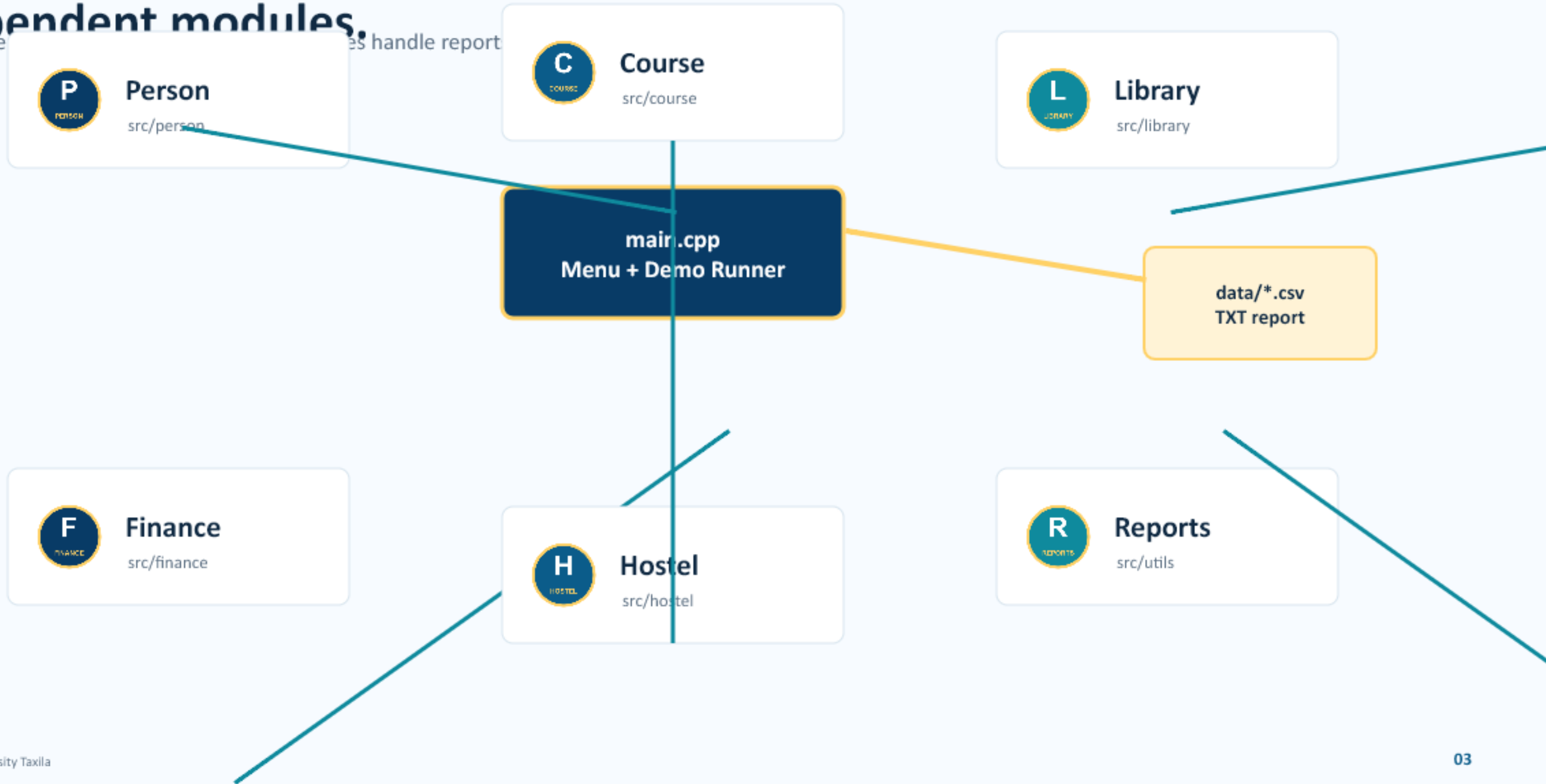
CSV

PERSISTENCE

A menu-driven console app coordinates six independent modules.

Each module

es handle report



The repository is organized by module, artifacts, and persisted data.

The structure mirrors the project checklist and keeps implementation boundaries clear.

src/

- └─ person
- └─ course
- └─ library
- └─ finance
- └─ hostel
- └─ utils

docs/

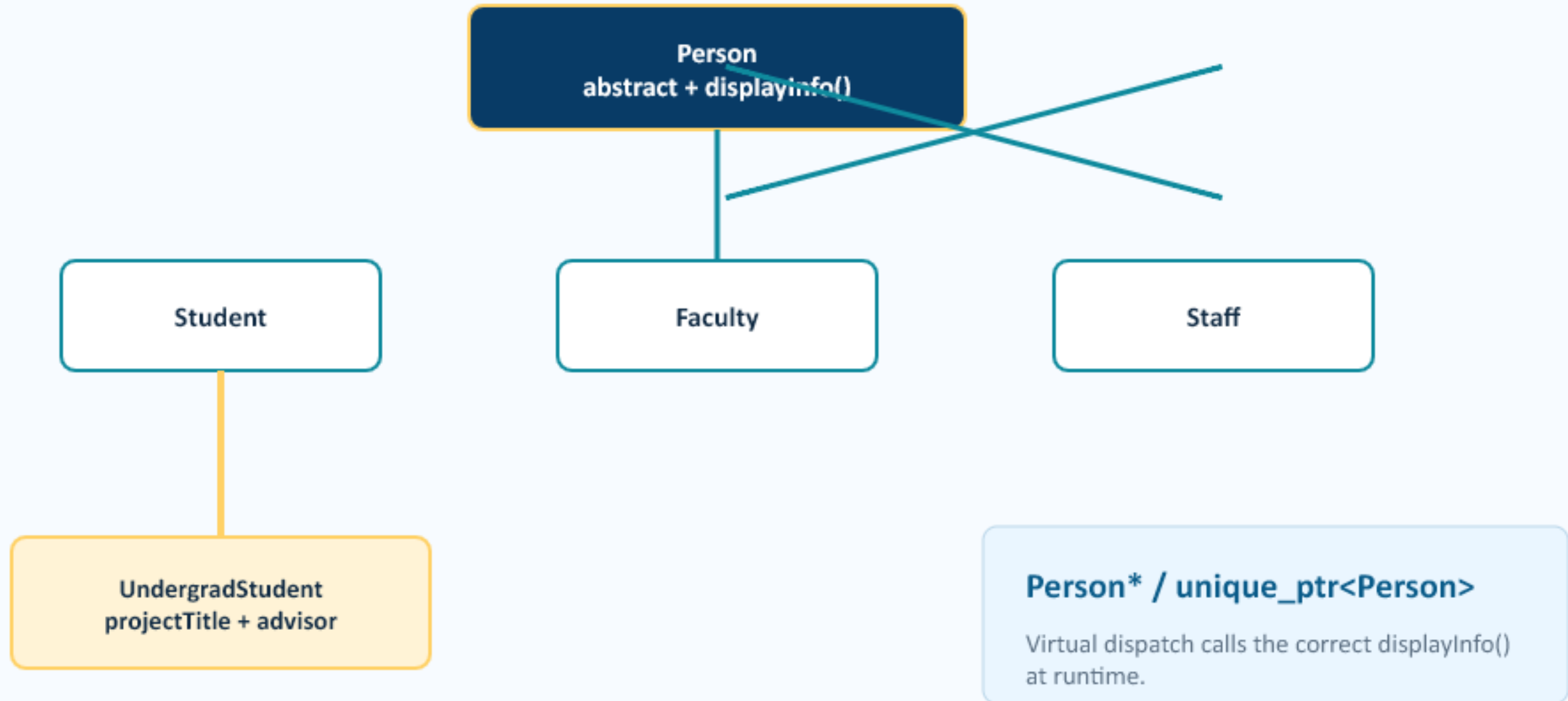
- └─ project docs
- └─ UML diagrams
- └─ logo + branding
- └─ cover page
- └─ presentation

data/

- └─ students.csv
- └─ courses.csv
- └─ library_items.csv
- └─ finance_summary.csv
- └─ hostel_rooms.csv
- └─ scms_report.txt

Runtime polymorphism starts from the abstract Person base class.

UndergradStudent preserves multilevel inheritance while keeping the student model accurate.



Enrollment rules are enforced through operators and custom exceptions.

Course capacity, credit hours, and enrollment data are validated before state changes.



+ **operator+=**

Enrolls a student into a course and increments enrolledStudents safely.

= **operator==**

Compares Course objects by courseCode for identity checks.

<< **operator<<**

Formats course details for clean console output.

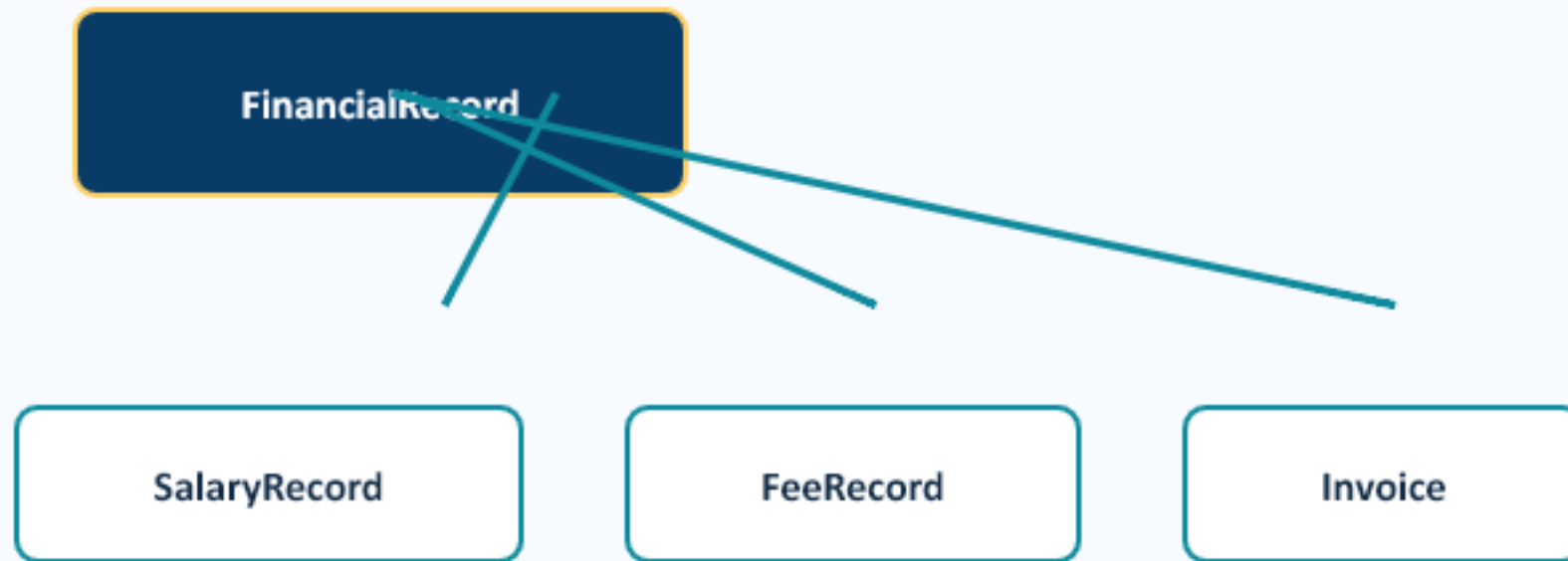
The library module combines abstraction, templates, and borrowing workflows.

Book and Journal objects are stored through LibraryItem polymorphism.



Financial records are validated, summarized, and persisted as CSV.

FinanceManager composes salary, fee, and invoice records.



Static members

SalaryRecord and FeeRecord maintain total record counters.

f Friend summary

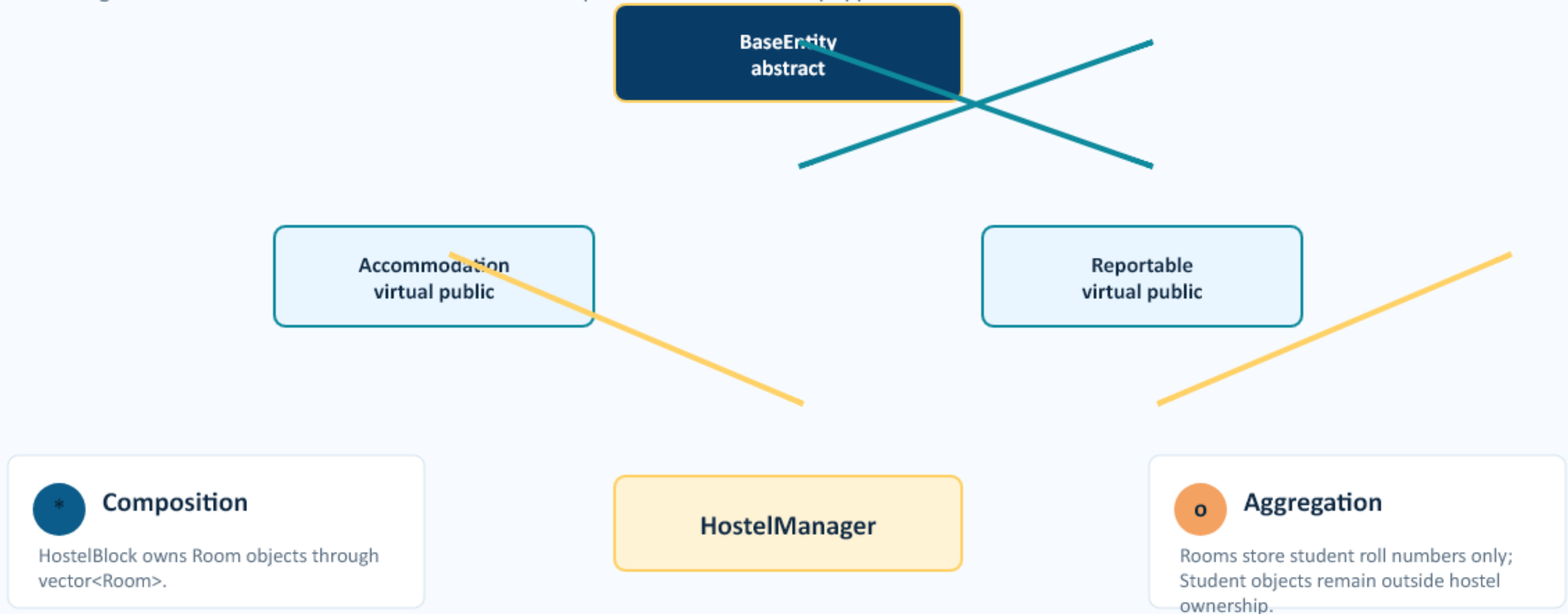
generateFinanceSummary() accesses private financial data for reports.

CSV Persistence

Finance CSV and invoice CSV save/load data through fstream RAI.

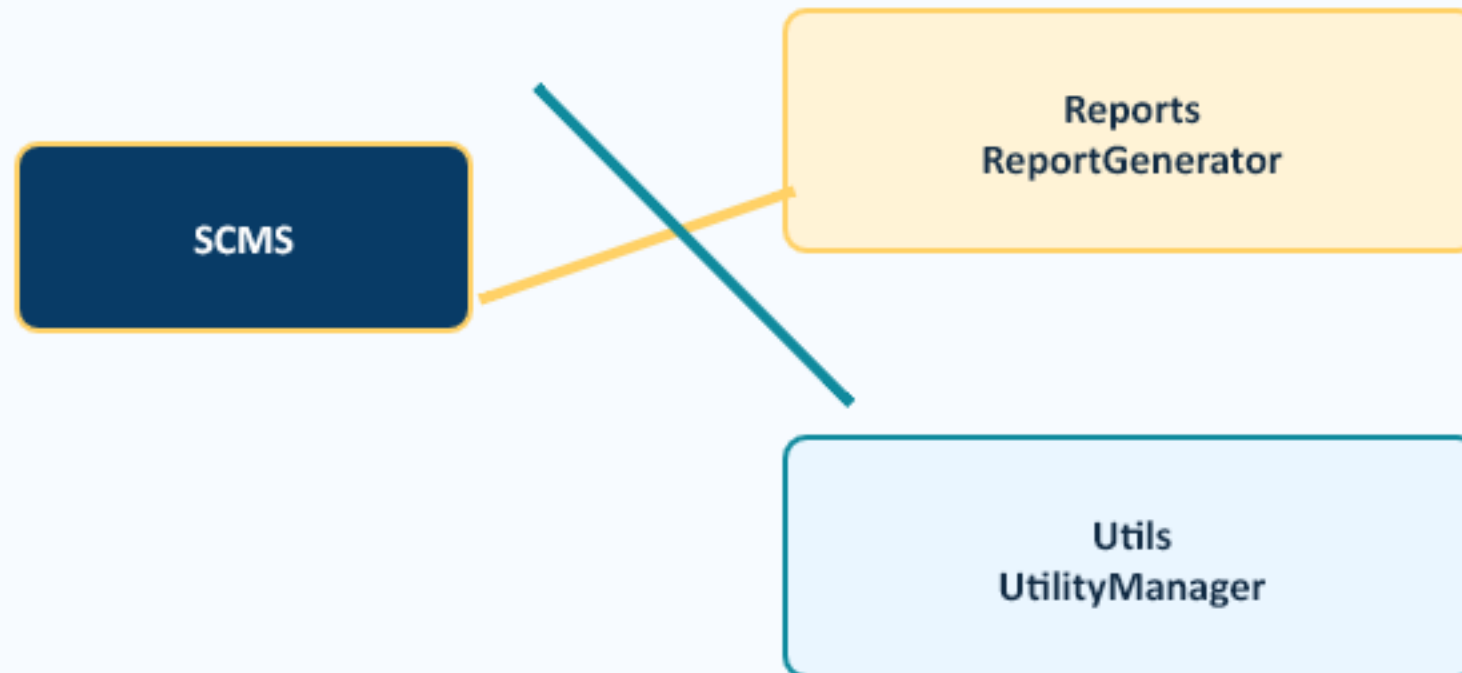
Virtual inheritance resolves the diamond problem cleanly.

HostelManager inherits behavior from Accommodation and Reportable while BaseEntity appears once.



Modern C++ features power campus-wide reporting.

Nested namespaces, smart pointers, lambdas, and STL algorithms organize shared functionality.



ptr Smart pointers

`unique_ptr<ReportGenerator>` and `shared_ptr<UtilityManager>` demonstrate safe ownership.

λ Lambdas

Sorting students, finding overdue library records, and filtering unpaid fees.

STL STL algorithms

`sort`, `find_if`, `count_if`, `for_each`, and `copy_if` are used meaningfully.

Every module saves and reloads data through structured files.

CSV files support module data; TXT output stores the generated campus report.



Safety rule

Bad CSV rows are skipped safely; numeric parsing validates before object creation.

students.csv
undergrad_students.csv
courses.csv
library_items.csv
finance_summary.csv
hostel_allocations.csv
scms_report.txt

The project covers the full instructor checklist with concrete classes.

Concepts are distributed across modules instead of being isolated examples.

Inheritance

Person, LibraryItem, FinancialRecord, Hostel

Polymorphism

displayInfo(), displayRecord(), LibraryItem*

Operators

+=, +, ==, <<, -=

Templates

Library<T>, searchByTitle<T>()

Exceptions

Enrollment, capacity, overdue

Smart pointers

unique_ptr, shared_ptr

Namespaces

SCMS::Reports, SCMS::Utils

Persistence

fstream CSV/TXT save/load

Composition

FinanceManager, HostelBlock, LibraryManager

The class model shows inheritance, composition, aggregation, and templates.

The full UML artifact is maintained in docs/UML and reused here for presentation.

SCMS UML Class Diagram

```

Person
|-- Student
|   |-- UndergraduateStudent
|-- Faculty
|-- Staff

Course o-- Faculty
Enrollment o-- Student

LibraryItem
|-- Book
|-- Journal
LibraryManager *-- LibraryItem
LibraryManager *-- LibraryMember
Library<T> template stores shared ptr<T>

FinancialRecord
|-- SalaryRecord
|-- FeeRecord
|-- Invoice
FinancialManager *-- SalaryRecord / FeeRecord / Invoice

BaseEntity
|-- Accommodation : virtual public
|-- Reportable : virtual public
Accommodation <|-- HostelManager
Reportable <|-- HostelManager
HostelManager *-- HostelBlock
HostelBlock *-- Room
Room o-- Student roll numbers

SCMS::Reports::ReportGenerator
SCMS::Utils::UtilityManager
  
```

I Inheritance

Person, LibraryItem, FinancialRecord, and BaseEntity hierarchies.

* Composition

Manager classes own their records, items, blocks, and rooms.

o Aggregation

Courses, rooms, and enrollments reference external student/faculty data.

The final project compiles cleanly and runs through the integrated menu.

Verification confirms executable generation, runtime startup, and module smoke paths.

```
$sources = Get-ChildItem src -Recurse -Filter *.cpp  
g++ -std=c++17 -Wall -Wextra -Werror $sources.FullName -o scms.exe
```



Smoke test path

Startup splash -> main menu -> selected modules -> return menu -> exit code 0.

PASS

ZERO WARNINGS

PASS

RUNTIME MENU

YES

SCMS.EXE

17

SLIDES

SCMS is submission-ready as a complete C++ OOP showcase.

The implementation combines academic requirements with practical campus workflows.

1 Complete modules

Person, Course, Library, Finance, Hostel, Reports

2 Strong OOP coverage

25+ concepts demonstrated in real classes

3 Persistence

CSV/TXT data files generated and reloaded

4 Documentation

Overview, architecture, UML, viva prep, testing

5 Branding

Logo, cover page, presentation, assets

6 Packaging

Executable and ZIP submission artifacts

A modular campus system that proves OOP through working software.

SCMS connects class design, runtime behavior, persistence, reporting, and polished submission assets.

A

Academic value

Every major concept is tied to a visible project feature and viva-ready explanation.

T

Technical value

C++17, STL, RAII, validation, exceptions, and clean module boundaries.

P

Presentation value

Professional documentation, UML, branding, and verified build output.

Ready for instructor evaluation and viva discussion.



Questions & Answers

Smart Campus Management System (SCMS) | HITEC University Taxila